INCH-POUND
MIL-DTL-24787/1A(SH)
24 September 2020
SUPERSEDING
MIL-F-24787/1(SH)
3 September 1993

DETAIL SPECIFICATION SHEET

FITTINGS, END, TYPES F, FL, AND FFL (FLANGE), REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24787.

1. SCOPE

1.1 Scope. This specification covers reusable flanged end fittings (types F, FL, and FFL) for hose sizes ranging from -12 to -192 for use with wire- or synthetic fiber-reinforced rubber, flexible hose assemblies in accordance with MIL-DTL-24135 and MIL-DTL-24136.

1.2 Classification. Flanged end fittings are of the sizes and constructions type F (straight nipple), type FL (90-degree elbow nipple), and type FFL (45-degree elbow nipple) as identified on figures 1 through 6. For the corresponding hose, see table I.

1.2.1 Classes. Flanges are of the following classes, as specified below and in accordance with figure 7 (see 6.2).

Class	Description
I	400-psi flange conforming to MIL-PRF-20042
II	150-psi flange conforming to MIL-PRF-20042 (with 400-psi flange thickness)
III	250-psi flange conforming to MIL-PRF-20042
IV	150-psi flange conforming to MIL-PRF-20042
V	150-psi flange conforming to ASME B16.5
VI	6,000-psi (42-megapascal) solid flange conforming to SAE J518-2

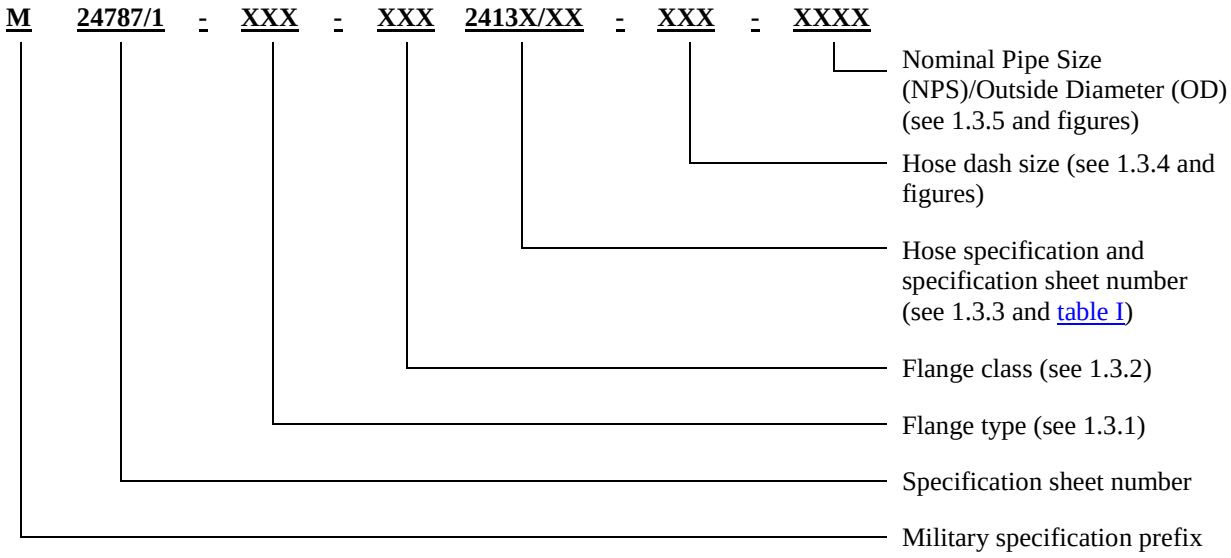
Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4730

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.

1.3 Part or identifying number (PIN). The PIN is a definitive part number that is formed by combining the applicable specification sheet number (M24787/1 for MIL-DTL-24787/1), a dash, flange type, a dash, flange class, hose specification and applicable specification sheet number, a dash, hose dash size, a dash, and nominal pipe size (NPS).



PIN example: M24787/1-FXX-IVO24135/01-32X-0150 describes a MIL-DTL-24787/1 end fitting, type F (straight nipple), flange class IV, for use on MIL-DTL-24135/1, MIL-DTL-24135/5, and MIL-DTL-24135/13 hoses, with a hose size of -32, and an NPS of 1½ inches.

1.3.1 Flange type. Flange types are identified in the PIN as follows:

Flange type	Description	PIN code
F	Straight nipple	FXX
FL	90-degree elbow nipple	FLX
FFL	45-degree elbow nipple	FFL

1.3.2 Flange class. Flange classes are identified in the PIN as follows:

Flange class	PIN code
I	IOO
II	IIO
III	III
IV	IVO
V	VOO
VI	VIO

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.

1.3.3 Hose specification and specification sheet number. Specification sheet numbers that only contain one digit will enter "0" as the first digit. For example, for specification sheet 1, the PIN code will be recorded as "01". For flanges categorized with multiple specification sheets, the first specification and applicable specification sheet will be used for the PIN. For example, for a flange that can be used with MIL-DTL-24135/1, MIL-DTL-24135/5, and MIL-DTL-24135/13, the PIN code will be recorded as "24135/01".

1.3.4 Hose dash size. Hose dash sizes in the tables of figures 1 through 6 that only contain two digits will enter "X" as the last digits. For example, for a hose dash size of -16, the PIN code will be recorded as "16X". For hose dash sizes that contain three digits, all digits will be entered in the PIN. For example, for a hose dash size of -128, the PIN code will be recorded as "128".

1.3.5 NPS/OD code. When the NPS/OD size in the tables of figures 1 through 4 is a value that does not contain a fraction, the inch size will be recorded in the first two digits followed by two "X's". For inch sizes less than 10, the first digit will be a zero. For example, for a size of 3 inches, the PIN code will be recorded as "03XX". For NPS/OD sizes that only contain a dimension with a fraction (e.g., "½"), the PIN code will be entered as identified below. When the NPS/OD size is a value of 1 inch or greater and contains a fraction, the "XX's" below will be replaced with the inch value. For example, for an NPS/OD size of 1½ inches, the PIN code will be recorded as "0150".

NPS/OD size	PIN code
¼	XX25
½	XX50
¾	XX75

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-T-1368 - Tube and Pipe, Nickel-Copper Alloy, Seamless and Welded

MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)

MIL-PRF-20042 - Flanges, Pipe, Bronze (Silver Brazing)

MIL-DTL-24135 - Hose, Synthetic Rubber, Wire Reinforced, for Flexible Hose Assemblies, General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.

MIL-DTL-24135/1	-	Hose, Synthetic Rubber, Wire Reinforced (Double-Wire Braid or Double-Spiral and Single-Wire Braid), for Flexible Hose Assemblies (High Pressure, Sizes -4 through -32)
MIL-DTL-24135/4	-	Hose, Synthetic Rubber, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (Medium Pressure, Sizes -40, -48, and -64)
MIL-DTL-24135/5	-	Hose, Synthetic Rubber, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (Perforated Cover, for High Pressure Gas, Sizes -4 through -32)
MIL-DTL-24135/9	-	Hose, Synthetic Rubber, Wire Reinforced (Four-Ply, Heavy Spiral-Wrapped), for Flexible Hose Assemblies (High Pressure, Sizes -8 through -32)
MIL-DTL-24135/10	-	Hose, Synthetic Rubber, Wire Reinforced (Single- or Double-Wire Braid), for Flexible Hose Assemblies (Medium Pressure, Sizes -4 through -48)
MIL-DTL-24135/12	-	Hose, Synthetic Rubber, High Performance, Wire Reinforced (Single-Wire Braid Over Single Synthetic Fiber Braid), for Flexible Hose Assemblies (Medium Pressure, High Temperature, Multi-Fluid, Sizes -4 through -40)
MIL-DTL-24135/13	-	Hose, Synthetic Rubber, High Performance, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (High Pressure, High Temperature, Multi-Based Oil, Sizes -4 through -32)
MIL-DTL-24136	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies, General Specification for
MIL-DTL-24136/1	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -80 through -192)
MIL-DTL-24136/3	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -4 through -32)
MIL-DTL-24136/4	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -40, -48, and -64)
MIL-B-24480	-	Bronze, Nickel-Aluminum (UNS No. C95800) Casting for Sea Water Service
MIL-C-24723	-	Castings, Nickel-Copper Alloy
MIL-DTL-24787	-	Fittings, End, Reusable, for Flexible Hose Assemblies, General Specification for

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 - Pipe Flanges and Flanged Fittings NPS ½ through NPS 24 Metric/Inch Standard

(Copies of this document are available online at http://webstore.ansi.org/.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.

ASTM INTERNATIONAL

ASTM A220/A220M	-	Standard Specification for Pearlitic Malleable Iron
ASTM A494/A494M	-	Standard Specification for Castings, Nickel and Nickel Alloy
ASTM B271/B271M	-	Standard Specification for Copper-Based Alloy Centrifugal Castings
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B505/B505M	-	Standard Specification for Copper-Based Alloy Continuous Castings
ASTM B582	-	Standard Specification for Nickel-Chromium-Iron-Molybdenum-Copper Alloy Plate, Sheet, and Strip
ASTM B584	-	Standard Specification for Copper Alloy Sand Castings for General Applications

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS6381	-	Steel, Mechanical Tubing 0.95Cr – 0.20Mo (0.38 – 0.43C) (SAE 4140)
SAE J518-2	-	Hydraulic Flanged Tube, Pipe, and Hose Connections, 4-Screw Flange Connection, Part 2: 42 MPa (Code 62)

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Flanged end fittings furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Flanged end fittings. Flanged end fittings shall be in accordance with MIL-DTL-24787 and as specified herein.

3.2.1 Dimensions. Flange dimensions shall be in accordance with figure 7. Straight nipples for type F flanged end fittings, 45-degree nipples for type FFL flanged end fittings, and 90-degree nipples for type FL flanged end fittings shall be in accordance with figures 1 through 6. All dimensions in figures 1 through 7 are recorded in inches.

3.2.2 Materials. End fitting materials shall be in accordance with figures 1 through 6. Nickel-based investment castings, if used, shall be in accordance with the material requirements of ASTM A494/A494M, with the exception of M-25S, and the welding and non-destructive testing (NDT) requirements of MIL-C-24723. M-25S made in accordance with ASTM A494/A494M shall be in accordance with MIL-C-24723 for mechanical property, welding, and NDT requirements.

3.2.3 Pressures. Flange pressures shall be as specified in 1.2.

3.2.4 End fitting availability. Flanged end fittings (types F, FL, and FFL) are available in the hoses and sizes indicated in table I (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.

TABLE I. Flanged end fitting by hose specification sheet and size.

End fitting size (hose dash number)	Hose specification sheet					
	24135/1, /5, and /13 (figure 1)	24135/4 and /10 (-40 and -48 only) (figure 2)	24135/9 (figure 3)	24135/10 (-4 to -32), /12, and 24136/3 (figure 4)	24136/1 (figure 5)	24136/4 (figure 6)
-12	X	--	--	--	--	--
-16	X	--	--	X	--	--
-20	X	--	X	X	--	--
-24	X	--	X	X	--	--
-32	X	--	X	X	--	--
-40	--	X	--	--	--	X
-48	--	X	--	--	--	X
-64	--	X	--	--	--	X
-80	--	--	--	--	X	--
-96	--	--	--	--	X	--
-128	--	--	--	--	X	--
-160	--	--	--	--	X	--
-192	--	--	--	--	X	--
NOTE: X denotes where flanged end fittings are available by hose specification sheet and size.						

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-DTL-24787. Flanged end fittings, because of their relatively low pressure ratings when compared to the rated pressures of the hoses upon which they are installed, shall not be subjected to qualification testing. They shall be considered qualified when other end fittings of the same hose specification sheet grouping (see table I) are qualified and those other fittings are not less than the rated pressure of the applicable hose.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-DTL-24787.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. End fittings to this specification are intended for use in shipboard or shore to ship piping systems where pressures do not exceed the flange rated working pressure and temperatures do not exceed that of the applicable hose.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of end fittings.

c. End fitting type, class, hose specifications, and size (hose by pipe or tube) (see 3.2.4).

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24787 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

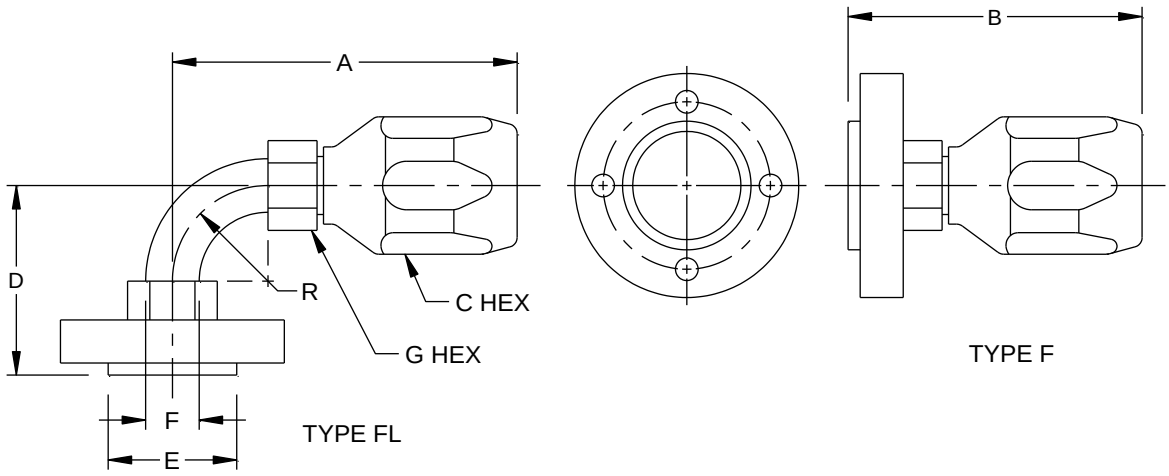
6.4 Subject term (key word) listing.

Synthetic fiber

Wire

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.



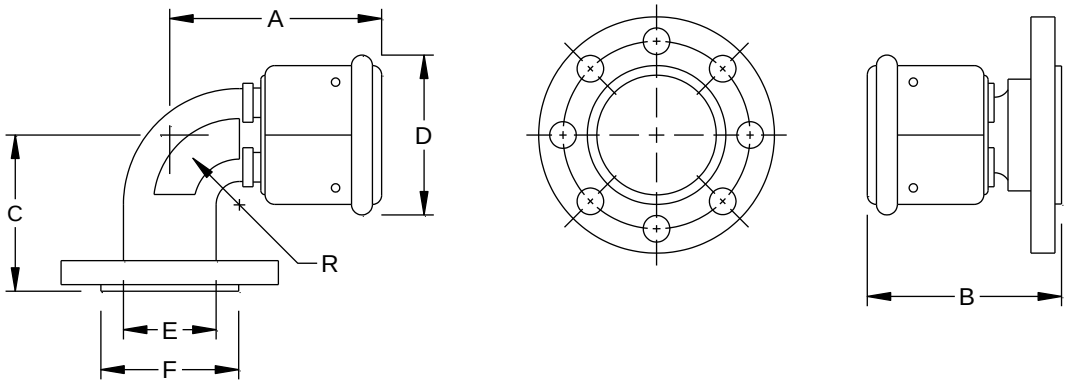
Hose size (dash number)	NPS	A nom	B nom	C nom	D nom	E	F	G	R
-12	½	4.52	3.92	1.50	2.19	1.50	0.71	1.12	1.25
-16	¾	5.25	4.34	1.81	2.49	1.84	0.92	1.38	1.50
-20	1	6.10	4.81	2.25	2.69	2.12	1.18	1.62	1.75
-24	1¼	6.31	4.97	2.50	3.13	2.74	1.53	2.12	2.00
-32	1½	7.38	5.33	3.00	3.00	2.74	1.77	2.25	2.50
-32	2	7.38	5.33	3.00	3.35	3.28	1.77	2.69	2.50

NOTES:

1. End fitting assembly materials for flange, shoulder, hose nipple, and socket shall be in accordance with QQ-N-281 or QQ-N-286.
2. The 90-degree tube elbow material for size a -12 end fitting shall be in accordance with MIL-T-1368 and for sizes -16 through -32, to MIL-T-16420.
3. Flange material alternative may be alloy C90300,in accordance with ASTM B271/B271M, ASTM B505/B505M, ASTM B584, or ASTM B61.
4. Socket (body) material alternative may be composition M-30H, in accordance with MIL-C-24723 or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
5. The flange shall be designed to swivel through 360 degrees and shall be dimensional in accordance with MIL-PRF-20042 and ASME B16.5 for the appropriate flange class.
6. The socket shall be a one-piece design.

FIGURE 1. MIL-DTL-24135/1, /5, and /13, flange to hose, type F (straight) and type FL (90-degree elbow).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.



TYPE FL TYPE F

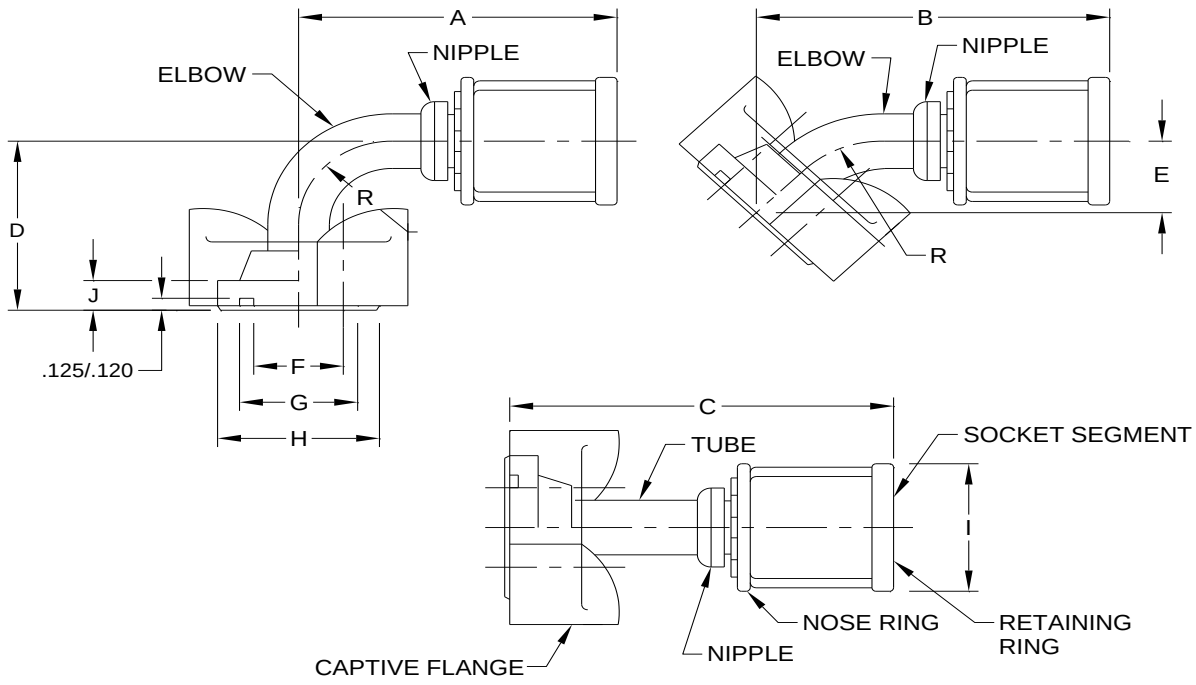
Hose size (dash number)	NPS	A nom	B nom	C nom	D ring and band nom	D bolted max	E	F	R
-40	2	5.70	5.16	3.50	3.90	6.75	2.24	3.28	1.88
-40	2½	5.70	5.16	3.50	3.90	6.75	2.24	3.28	1.88
-48	2½	5.96	5.17	4.00	4.50	7.75	2.71	3.78	2.09
-48	3	5.96	5.17	4.00	4.50	7.75	3.31	4.50	2.09
-64	3½	6.75	5.54	4.62	5.44	9.00	3.81	5.22	2.81
-64	4	6.75	5.54	4.62	5.44	9.00	4.28	5.22	2.81

NOTES:

1. End fitting assembly materials for flange, hose nipple, and socket assembly shall be in accordance with MIL-B-24480.
2. Socket assembly may be multiple segment type, bolt, or ring retained, so long as the envelope dimensions are not exceeded.
3. Socket assembly material alternative may be composition M-30H in accordance with QQ-N-281, class A or B, QQ-N-286, or MIL-C-24723 or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
4. Flange material alternative may be alloy C90300, in accordance with ASTM B271/B271M, ASTM B505/B505M, ASTM B584, or ASTM B61.
5. Flange shall be designed to swivel through 360 degrees and shall be dimensioned in accordance with MIL-PRF-20042 and ASME B16.5 for the appropriate flange class.
6. The “D” dimension represents dimensions for a ring and band fitting or for a bolted fitting of this type.

FIGURE 2. MIL-DTL-24135/4 and /10 (-40 and -48 only), flange to hose, type F (straight) and type FL (90-degree elbow).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.



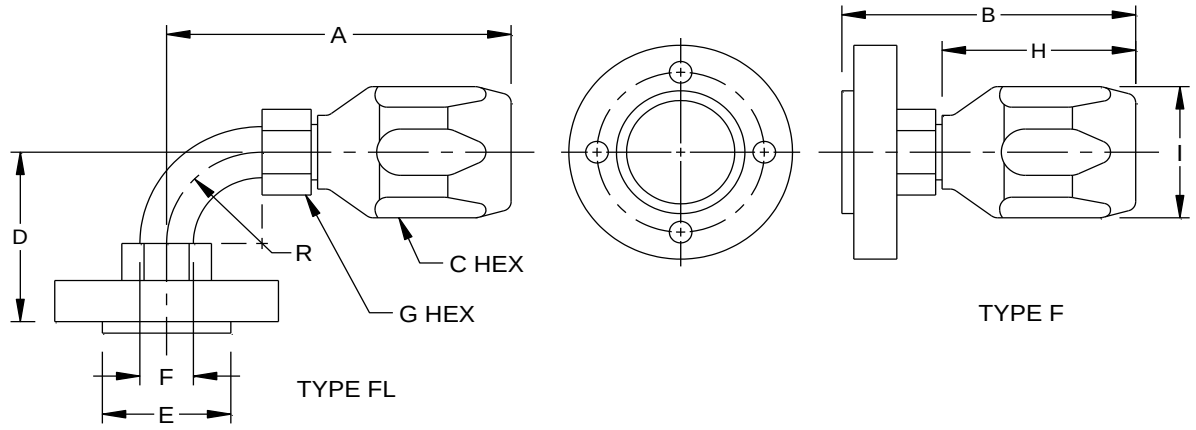
Size		A max	B max	C max	D nom	E nom	F nom	G nom	H nom	I max	J nom	R
OD tube	SAE flange & hose											
1¼	-20	6.44	7.34	6.68	2.68	1.96	1.422	1.752	2.125	2.88	0.405	1.75
1½	-24	7.25	7.65	7.14	3.19	2.06	1.790	2.120	2.500	3.16	0.495	2.00
2	-32	8.95	8.94	9.33	4.56	3.31	2.165	2.495	3.125	3.75	0.495	3.00

NOTES:

1. End fitting assembly materials for the hose nipple and socket shall be in accordance with QQ-N-281 or QQ-N-286.
2. Socket for end fitting assemblies for sizes -20 hose and larger may be the one-piece or multiple-segment design, not exceeding the envelope dimensions.
3. End fitting assembly alternative materials for the socket may be composition M-30H in accordance with QQ-N-281, QQ-N-286, or MIL-C-24723 or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
4. Alternate material for the end fitting assembly socket for hose sizes -20 and larger maybe be AISI/SAE-4140 steel in accordance with SAE AMS6381, as applicable.
5. Tubing shall be in accordance with MIL-T-16420.
6. Captive flange material shall be malleable iron in accordance with ASTM A220/A220M.

FIGURE 3. MIL-DTL-24135/9, flange to hose, type F (straight), type FL (90-degree elbow), and type FFL (45-degree elbow), SAE J518-2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.



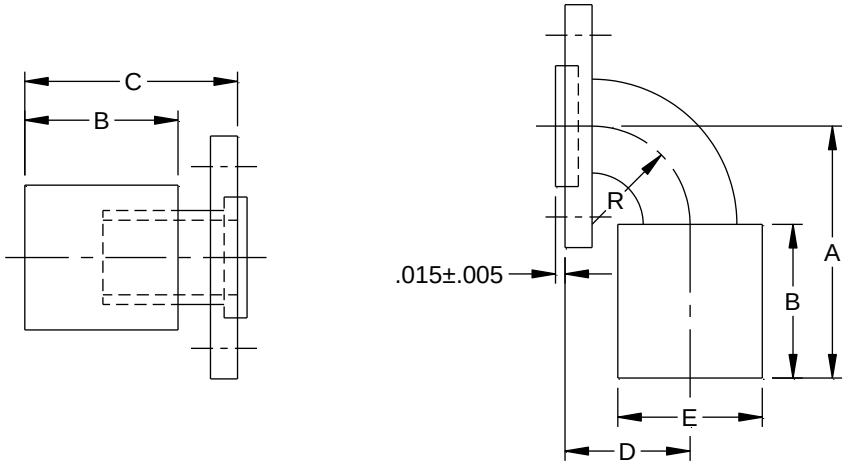
Hose size (dash number)	NPS	A nom	B nom	C nom	D nom	E nom	F nom	G	H	I	R
									See note 6		
-16	½	2.80	2.80	1.44	2.38	1.50	0.71	1.38	1.66	1.53	1.12
-16	¾	2.80	2.80	1.44	2.38	1.50	0.71	1.38	1.66	1.53	1.12
-16	1	2.80	2.80	1.44	2.38	1.50	0.71	1.38	1.66	1.53	1.12
-20	¾	2.98	2.89	1.75	2.56	1.84	0.92	1.62	1.75	1.62	1.22
-24	1	3.24	2.92	1.94	2.75	2.12	1.18	2.12	1.84	1.78	1.38
-32	1¼	3.89	3.51	2.44	3.06	2.74	1.53	2.25	2.25	2.05	1.62
-32	1½	3.89	3.51	2.44	3.06	2.74	1.77	2.25	2.25	2.05	1.62
-32	2	3.89	3.51	2.44	3.06	2.74	1.75	2.69	2.25	2.05	1.62

NOTES:

1. End fitting assembly materials for flange, shoulder, hose nipple, and socket shall be in accordance with QQ-N-281 or QQ-N-286.
2. The 90-degree tube elbow material shall be in accordance with MIL-T-16420.
3. Alternative material for the flange may be alloy C90300 in accordance with ASTM B271/B271M, ASTM B505/B505M, ASTM B584, or ASTM B61.
4. Socket (body) material alternative may be composition M-30H in accordance with MIL-C-24723 or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance. Socket may be one-piece or segmented.
5. The flange shall be designed to swivel through 360 degrees and shall be dimensioned in accordance with MIL-PRF-20042 and ASME B16.5 for the appropriate flange class.
6. Dimensions H and I are nominal dimensions for bolt-together type segments.

FIGURE 4. MIL-DTL-24135/10 (-4 to -32), /12, and MIL-DTL-24136/3, flange to hose, type F (straight) and type FL (90-degree elbow).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.



TYPE F TYPE FL

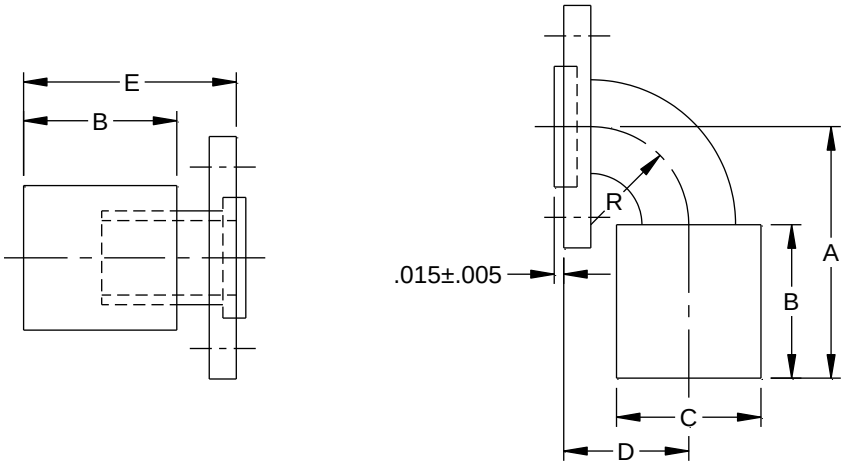
Hose size (dash number)	A	B	C nom	D	E	R
-80	8.64	5.00	6.75	6.81	11.50	3.50
-96	9.14	5.00	6.75	7.31	12.00	4.00
-128	10.34	5.00	7.00	8.81	14.00	5.00
-160	11.89	5.00	7.01	9.70	16.50	6.38
-192	14.24	6.50	8.49	10.70	20.00	7.38

NOTES:

1. Parts in contact with seawater shall be bronze in accordance with MIL-B-24480. All fasteners shall be in accordance with QQ-N-281 or QQ-N-286.
2. The flange shall be in accordance with MIL-B-24480 or ASTM B271/B271M, ASTM B505/B505M, ASTM B584, or ASTM B61.
3. The flange shall be designed to swivel through 360 degrees and shall be dimensioned in accordance with MIL-PRF-20042 and ASME B16.5 for the appropriate flange class.

FIGURE 5. MIL-DTL-24136/1, flange to hose, type F (straight) and type FL (90-degree elbow).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.



TYPE F TYPE FL

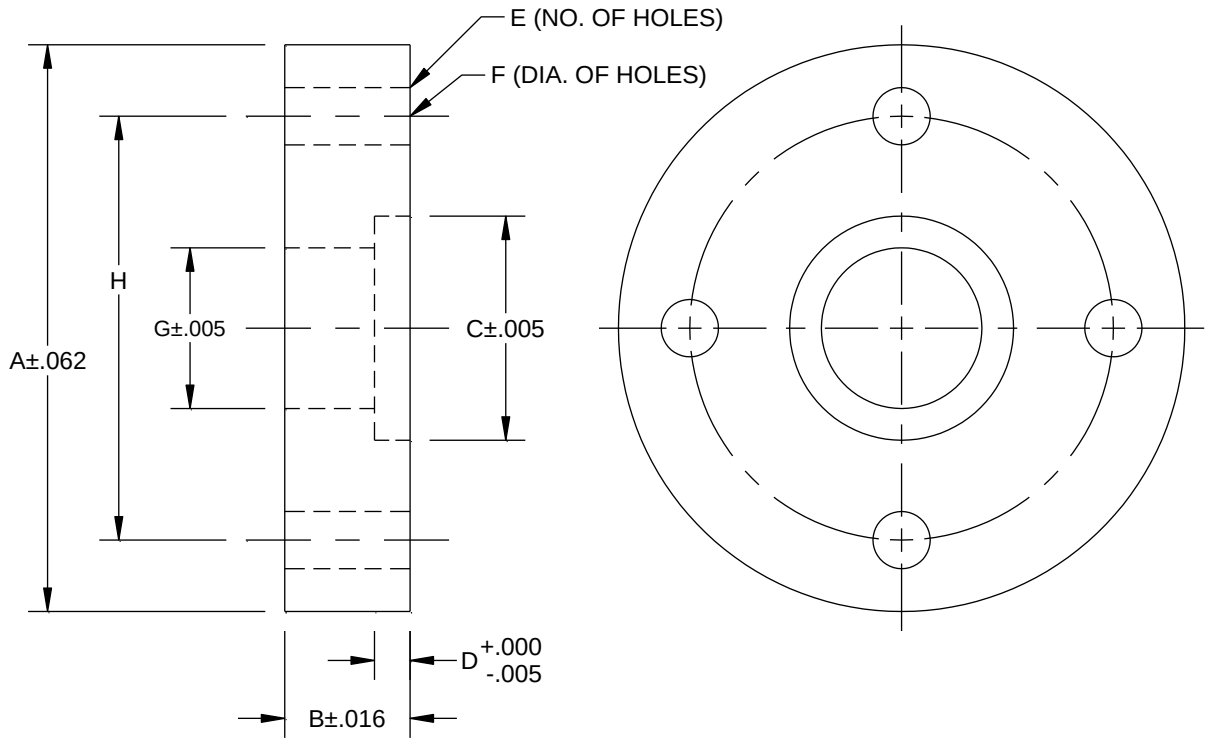
Hose size (dash no.)	A nom	B	C	D	E nom	R
-40	6.25	5.00	6.75	4.50	5.75	2.13
-48	6.88	5.00	7.75	5.00	5.88	2.50
-64	7.00	5.00	9.00	5.57	6.00	3.00

NOTES:

1. Parts in contact with seawater shall be bronze in accordance with MIL-B-24480. All fasteners shall be in accordance with QQ-N-281 or QQ-N-286.
2. The flange shall be in accordance with MIL-B-24480 or ASTM B271/B271M, ASTM B505/B505M, ASTM B584, or ASTM B61.
3. Flange shall be designed to swivel through 360 degrees and shall be dimensioned in accordance with MIL-PRF-20042 and ASME B16.5for the appropriate flange class.

FIGURE 6. MIL-DTL-24136/4, flange to hose, type F (straight) and type FL (90-degree elbow).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.



NOTES:

1. See MIL-PRF-20042 and ASME B16.5 for detail dimensions.
2. Flanges in accordance with MIL-PRF-20042 class II, 150-psi flange (with 400-psi flange thickness) are for replacement purposes only and shall not be used in new construction.
3. Flange dimension identification:

A	Outside diameter
B	Height
C	Countersunk bore diameter
D	Countersunk bore height
E	Number of bolt holes
F	Diameter of bolt holes
G	Inside diameter
H	Diameter of bolt circle

FIGURE 7. Flange classes. (see 1.2.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Preparing activity:
Navy − SH
(Project 4730-2019-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:38Z
Check the source to verify that this is the current version before use.